

QUESTIONS

Q1
Student Management System
(BL3 - Apply)
10 marks

Q2
2. Library Book Management
(BL3 - Apply)
10 marks

Q3
Employee Salary Calculation
(BL3 - Apply)
10 marks

Q4
Shape Area Calculator using
Polymorphism (BL3 - Apply)
10 marks

Q5
Vehicle Registration System
(BL3 - Apply)
10 marks

Q6
Bank Account Hierarchy (BL4 -
Analyze)
10 marks

Q7
Hospital Patient Record System
(BL4 - Analyze)
10 marks

Q8
E-commerce Product Catalog
(BL4 - Analyze)
10 marks

Q9
University Staff Hierarchy (BL4 -
Analyze)
10 marks

Q10
Smart Home Device Controller
(BL4 - Analyze)
10 marks

10/10 answered

Q1 Student Management System (BL3 - Apply)

10 marks

Write pseudocode to design a Student class with attributes (name, rollNo, marks) and methods to:

- Accept student details
- Calculate average marks
- Display result (Pass/Fail)

Apply encapsulation and data hiding by restricting direct access to attributes.

OOP Concepts: Encapsulation | Data Hiding | Classes & Objects

Your Answer

```

CLASS Student
    PRIVATE String name
    PRIVATE String rollNo
    PRIVATE Float mark1, mark2, mark3
    PRIVATE Float average
    PUBLIC METHOD processStudentDetails()
        // 1. Accept Details
        PRINT "Enter Student Name:"
        READ name
        PRINT "Enter Roll / Registration Number:"
        READ rollNo
        PRINT "Enter Marks for 3 Subjects:"
        READ mark1, mark2, mark3
        // 2. Calculate Average
        SET average = (mark1 + mark2 + mark3) / 3.0
        // 3. Display Result
        PRINT "-----"
        PRINT "Student Details & Result"
        PRINT "-----"
        PRINT "Name: " + name
        PRINT "Roll No: " + rollNo
        PRINT "Average Marks: " + average
        IF average >= 50 THEN
            PRINT "Result: PASS"
        ELSE
            PRINT "Result: FAIL"
        END IF
    END METHOD
END CLASS

// Program Execution
START
    CREATE studentObj = NEW Student()
    studentObj.processStudentDetails()
END
    
```

Save Answer

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10/10 answered

Q2 2. Library Book Management [BL3 – Apply]

10 marks

Design a Book class with private attributes (title, author, ISBN, isAvailable). Implement methods to:

- Issue a book (change availability status)
- Return a book
- Display book details using getter methods

Demonstrate object creation for at least 3 books and simulate issue/return operations.

OOP Concepts: Encapsulation | Getters & Setters | Objects

Your Answer

```

CLASS Book
    PRIVATE String title
    PRIVATE String author
    PRIVATE String isbn
    PRIVATE Boolean isAvailable
    CONSTRUCTOR Book(bTitle, bAuthor, bisbn)
        SET title = bTitle
        SET author = bAuthor
        SET isbn = bisbn
        SET isAvailable = TRUE
    END CONSTRUCTOR
    // Getter methods
    PUBLIC METHOD getTitle() RETURNS String
        RETURN title
    END METHOD
    PUBLIC METHOD getAuthor() RETURNS String
        RETURN author
    END METHOD
    PUBLIC METHOD getIsbn() RETURNS String
        RETURN isbn
    END METHOD
    PUBLIC METHOD getIsAvailable() RETURNS Boolean
        RETURN isAvailable
    END METHOD
    // Method to issue a book
    PUBLIC METHOD issueBook()
        IF isAvailable == TRUE THEN
            SET isAvailable = FALSE
            PRINT "Book "" + title + "" issued successfully."
        ELSE
            PRINT "Book "" + title + "" is currently unavailable."
        END IF
    END METHOD
    // Method to return a book
    PUBLIC METHOD returnBook()
        IF isAvailable == FALSE THEN
            SET isAvailable = TRUE
            PRINT "Book "" + title + "" returned successfully."
        ELSE
            PRINT "Book "" + title + "" was not issued."
        
```

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10/10 answered

Q3 Employee Salary Calculation [BL3 - Apply]

10 marks

Write pseudocode for an Employee class with encapsulated attributes (empId, name, basicPay, designation). Implement methods to:

- Calculate gross salary (basic + HRA + DA)
- Calculate net salary after deductions (PF, tax)
- Display pay slip

Apply data hiding so salary components are not directly accessible.
OOP Concepts: Encapsulation | Data Hiding | Methods

Your Answer

```

CLASS Employee
    PRIVATE String empId
    PRIVATE String name
    PRIVATE String designation
    PRIVATE Float basicPay
    CONSTRUCTOR Employee(eld, eName, eDesignation, eBasic)
        SET empId = eld
        SET name = eName
        SET designation = eDesignation
        SET basicPay = eBasic
    END CONSTRUCTOR
    PRIVATE METHOD calculateHRA() RETURNS Float
        RETURN basicPay * 0.20 // 20% HRA
    END METHOD
    PRIVATE METHOD calculateDA() RETURNS Float
        RETURN basicPay * 0.10 // 10% DA
    END METHOD
    PRIVATE METHOD calculatePF() RETURNS Float
        RETURN basicPay * 0.12 // 12% PF deduction
    END METHOD
    PRIVATE METHOD calculateTax(grossSalary) RETURNS Float
        IF grossSalary > 50000 THEN
            RETURN grossSalary * 0.10 // 10% tax for higher gross
        ELSE
            RETURN grossSalary * 0.05 // 5% tax
        END IF
    END METHOD
    // Public method to calculate gross salary (Basic + HRA + DA)
    PUBLIC METHOD calculateGrossSalary() RETURNS Float
        SET hra = calculateHRA()
        SET da = calculateDA()
        RETURN basicPay + hra + da
    END METHOD
    // Public method to calculate net salary (Gross - Deductions)
    PUBLIC METHOD calculateNetSalary() RETURNS Float
        SET gross = calculateGrossSalary()
        SET pf = calculatePF()
        SET tax = calculateTax(gross)
        SET totalDeductions = pf + tax
        RETURN gross - totalDeductions
    END METHOD
    
```

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Q4 Shape Area Calculator using Polymorphism [BL3 - Apply]

10 marks

Create a Shape base class with an overridable calculateArea() method. Apply this to:

- Circle — area using radius
- Rectangle — area using length and breadth
- Triangle — area using base and height

Use method overriding to demonstrate runtime polymorphism. Call calculateArea() for each object.

OOP Concepts: Inheritance | Polymorphism | Method Overriding

Your Answer

```
PUBLIC METHOD calculateArea() RETURNS Float
    RETURN 0.0
END METHOD
END CLASS

CLASS Circle EXTENDS Shape
    PRIVATE Float radius
    CONSTRUCTOR Circle(r)
        SET radius = r
    END CONSTRUCTOR
    OVERRIDE PUBLIC METHOD calculateArea() RETURNS Float
        RETURN 3.14159 * radius * radius
    END METHOD
END CLASS

CLASS Rectangle EXTENDS Shape
    PRIVATE Float length
    PRIVATE Float breadth
    CONSTRUCTOR Rectangle(l, b)
        SET length = l
        SET breadth = b
    END CONSTRUCTOR
    OVERRIDE PUBLIC METHOD calculateArea() RETURNS Float
        RETURN length * breadth
    END METHOD
END CLASS

CLASS Triangle EXTENDS Shape
    PRIVATE Float base
    PRIVATE Float height
    CONSTRUCTOR Triangle(b, h)
        SET base = b
        SET height = h
    END CONSTRUCTOR
    OVERRIDE PUBLIC METHOD calculateArea() RETURNS Float
        RETURN 0.5 * base * height
    END METHOD
END CLASS

START
    CREATE shape1 = NEW Circle(5.0)
    CREATE shape2 = NEW Rectangle(4.0, 6.0)
    CREATE shape3 = NEW Triangle(3.0, 8.0)
    // Array of base class references (Polymorphism)
    CREATE Array of Shape shapes = [shape1, shape2, shape3]
    PRINT "=== SHAPE AREA CALCULATIONS ==="
    FOR EACH shape IN shapes DO
        PRINT "Area of " + shape.calculateArea()
```

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Q5 Vehicle Registration System [BL3 - Apply]

10 marks

Design a Vehicle class with attributes (regNo, owner, fuelType). Extend it into:

- TwoWheeler — with engine capacity
- FourWheeler — with seating capacity and AC status

Apply inheritance to reuse common attributes and override a displayDetails() method in each subclass.

OOP Concepts: Inheritance | Method Overriding | Subclass

Your Answer

```

CLASS Vehicle
    PROTECTED String regNo
    PROTECTED String owner
    PROTECTED String fuelType
    CONSTRUCTOR Vehicle(vRegNo, vOwner, vFuelType)
        SET regNo = vRegNo
        SET owner = vOwner
        SET fuelType = vFuelType
    END CONSTRUCTOR
    PUBLIC METHOD displayDetails()
        PRINT "Registration Number: " + regNo
        PRINT "Owner Name      : " + owner
        PRINT "Fuel Type       : " + fuelType
    END METHOD
END CLASS

CLASS TwoWheeler EXTENDS Vehicle
    PRIVATE Integer engineCapacity // in CC
    CONSTRUCTOR TwoWheeler(vRegNo, vOwner, vFuelType, capacity)
        CALL SUPER(vRegNo, vOwner, vFuelType)
        SET engineCapacity = capacity
    END CONSTRUCTOR
    OVERRIDE PUBLIC METHOD displayDetails()
        PRINT "=== TWO WHEELER DETAILS ==="
        CALL SUPER.displayDetails()
        PRINT "Engine Capacity  : " + engineCapacity + " CC"
        PRINT "-----"
    END METHOD
END CLASS

CLASS FourWheeler EXTENDS Vehicle
    PRIVATE Integer seatingCapacity
    PRIVATE Boolean hasAC
    CONSTRUCTOR FourWheeler(vRegNo, vOwner, vFuelType, capacity, acStatus)
        CALL SUPER(vRegNo, vOwner, vFuelType)
        SET seatingCapacity = capacity
        SET hasAC = acStatus
    END CONSTRUCTOR
    OVERRIDE PUBLIC METHOD displayDetails()
        PRINT "=== FOUR WHEELER DETAILS ==="
        CALL SUPER.displayDetails()
        PRINT "Seating Capacity : " + seatingCapacity
        IF hasAC == TRUE THEN
            PRINT "AC Status      : Available"
        ELSE
            PRINT "AC Status      : Not Available"
        END IF
    END IF
END IF
    
```



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10/10 answered

Q6 Bank Account Hierarchy [BL4 – Analyze]

10 marks

Develop pseudocode for a BankAccount superclass and two subclasses — SavingsAccount (with interest calculation) and CurrentAccount (with overdraft facility). Analyze and implement:

- Inheritance of common attributes (accNo, balance, holder)
- Method overriding for withdrawal behavior
- How overdraft protection differs from savings limit

Compare the behavior of withdraw() across both subclasses and justify the design choices.

OOP Concepts: Inheritance | Method Overriding | Polymorphism

Your Answer

```

CLASS BankAccount
    PROTECTED String accNo
    PROTECTED String holder
    PROTECTED Float balance
    CONSTRUCTOR BankAccount(pAccNo, pHolder, initialBalance)
        SET accNo = pAccNo
        SET holder = pHolder
        SET balance = initialBalance
    END CONSTRUCTOR
    PUBLIC METHOD deposit(amount)
        IF amount > 0 THEN
            SET balance = balance + amount
            PRINT "Deposited: $" + amount + " | New Balance: $" + balance
        ELSE
            PRINT "Invalid deposit amount."
        END IF
    END METHOD
    PUBLIC METHOD withdraw(amount) RETURNS Boolean
        IF amount > 0 AND balance >= amount THEN
            SET balance = balance - amount
            PRINT "Withdrew: $" + amount + " | Remaining Balance: $" + balance
            RETURN TRUE
        ELSE
            PRINT "Transaction Failed: Insufficient funds."
            RETURN FALSE
        END IF
    END METHOD
    PUBLIC METHOD displayDetails()
        PRINT "Account No: " + accNo
        PRINT "Holder : " + holder
        PRINT "Balance : $" + balance
    END METHOD
END CLASS

CLASS SavingsAccount EXTENDS BankAccount
    PRIVATE Float interestRate // e.g., 0.05 for 5%
    CONSTRUCTOR SavingsAccount(pAccNo, pHolder, initialBalance, rate)
        CALL SUPER(pAccNo, pHolder, initialBalance)
        SET interestRate = rate
    END CONSTRUCTOR
    PUBLIC METHOD calculateInterest()

```

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- Method overriding for withdrawal behavior
- How overdraft protection differs from savings limit

Compare the behavior of withdraw() across both subclasses and justify the design choices.
OOP Concepts: Inheritance | Method Overriding | Polymorphism

Your Answer

```
PUBLIC METHOD calculateInterest()
    SET interest = balance * interestRate
    SET balance = balance + interest
    PRINT "Interest Earned: $" + interest + " | Updated Balance: $" + balance
END METHOD

OVERRIDE PUBLIC METHOD withdraw(amount) RETURNS Boolean
    IF amount > 0 AND balance >= amount THEN
        SET balance = balance - amount
        PRINT "[Savings] Withdrew: $" + amount + " | Remaining Balance: $" + balance
        RETURN TRUE
    ELSE
        PRINT "[Savings] Transaction Failed: Cannot withdraw beyond current balance."
        RETURN FALSE
    END IF
END METHOD

END CLASS

CLASS CurrentAccount EXTENDS BankAccount
    PRIVATE Float overdraftLimit // Maximum allowed negative balance
    CONSTRUCTOR CurrentAccount(pAccNo, pHolder, initialBalance, limit)
        CALL SUPER(pAccNo, pHolder, initialBalance)
        SET overdraftLimit = limit
    END CONSTRUCTOR
    OVERRIDE PUBLIC METHOD withdraw(amount) RETURNS Boolean
        IF amount > 0 AND (balance + overdraftLimit) >= amount THEN
            SET balance = balance - amount
            PRINT "[Current] Withdrew: $" + amount + " | Updated Balance: $" + balance
            RETURN TRUE
        ELSE
            PRINT "[Current] Transaction Failed: Overdraft limit exceeded!"
            RETURN FALSE
        END IF
    END METHOD
END CLASS

START
    CREATE savings = NEW SavingsAccount("SA-101", "Alice", 5000.0, 0.05)
    CREATE current = NEW CurrentAccount("CA-202", "Bob", 2000.0, 1000.0)

    PRINT "=== SAVINGS ACCOUNT ==="
    savings.displayDetails()
    savings.calculateInterest()
```

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Q7 Hospital Patient Record System [BL4 - Analyze]

10 marks

Design a Patient base class and analyze how it extends into:

- InPatient — with ward number, admission date, and daily charges
- OutPatient — with appointment date and consultation fee

Analyze encapsulation requirements for sensitive data (diagnosis, prescription). Override a generateBill() method for each type and compare the billing logic differences. OOP Concepts: Encapsulation | Inheritance | Polymorphism

Your Answer

```

CLASS Patient
    PROTECTED String patientId
    PROTECTED String name
    PROTECTED Integer age
    PRIVATE String diagnosis
    PRIVATE String prescription
    CONSTRUCTOR Patient(pId, pName, pAge, pDiagnosis, pPrescription)
        SET patientId = pId
        SET name = pName
        SET age = pAge
        SET diagnosis = pDiagnosis
        SET prescription = pPrescription
    END CONSTRUCTOR
    PUBLIC METHOD getDiagnosis() RETURNS String
        RETURN diagnosis
    END METHOD
    PUBLIC METHOD setDiagnosis(newDiagnosis)
        SET diagnosis = newDiagnosis
    END METHOD
    PUBLIC METHOD getPrescription() RETURNS String
        RETURN prescription
    END METHOD
    PUBLIC METHOD setPrescription(newPrescription)
        SET prescription = newPrescription
    END METHOD
    PUBLIC METHOD generateBill() RETURNS Float
        RETURN 0.0
    END METHOD
    PUBLIC METHOD displayPatientInfo()
        PRINT "Patient ID : " + patientId
        PRINT "Name      : " + name
        PRINT "Age       : " + age
        PRINT "Diagnosis : " + getDiagnosis()
        PRINT "Prescription: " + getPrescription()
    END METHOD
END CLASS

CLASS InPatient EXTENDS Patient
    PRIVATE String wardNumber
    PRIVATE String admissionDate
    PRIVATE Float dailyCharge
    PRIVATE Integer numberOfDays
    CONSTRUCTOR InPatient(pId, pName, pAge, pDiag, pPres, ward, admDate, charge, days)

```

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Your Answer

```

CALL SUPER(pId, pName, pAge, pDiag, pPres)
SET wardNumber = ward
SET admissionDate = admDate
SET dailyCharge = charge
SET numberOfDays = days
END CONSTRUCTOR
OVERRIDE PUBLIC METHOD generateBill() RETURNS Float
RETURN dailyCharge * numberOfDays
END METHOD
PUBLIC METHOD displayInPatientDetails()
CALL SUPER.displayPatientInfo()
PRINT "Type : InPatient"
PRINT "Ward No : " + wardNumber
PRINT "Admission : " + admissionDate
PRINT "Days Stayed: " + numberOfDays
PRINT "Total Bill : $" + generateBill()
PRINT "-----"
END METHOD
END CLASS

CLASS OutPatient EXTENDS Patient
PRIVATE String appointmentDate
PRIVATE Float consultationFee

CONSTRUCTOR OutPatient(pId, pName, pAge, pDiag, pPres, appDate, fee)
CALL SUPER(pId, pName, pAge, pDiag, pPres)
SET appointmentDate = appDate
SET consultationFee = fee
END CONSTRUCTOR
OVERRIDE PUBLIC METHOD generateBill() RETURNS Float
RETURN consultationFee
END METHOD
PUBLIC METHOD displayOutPatientDetails()
CALL SUPER.displayPatientInfo()
PRINT "Type : OutPatient"
PRINT "Appointment: " + appointmentDate
PRINT "Total Bill : $" + generateBill()
PRINT "-----"
END METHOD
END CLASS

START
CREATE inPatient = NEW InPatient("P101", "Alice", 35, "Pneumonia", "Antibiotics", "Ward-A3", "2026-07-20", 500.0, 4)

```

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10/10 answered

Q8 E-Commerce Product Catalog [BL4 - Analyze]

10 marks

Analyze and design a Product class hierarchy for an e-commerce system:

- Electronics — with warranty, brand, voltage
- Clothing — with size, fabric, gender
- Grocery — with expiryDate and weight

Override an applyDiscount() method for each category with different discount rules. Analyze why polymorphism makes the cart checkout process flexible and extensible. OOP Concepts: Polymorphism | Inheritance | Method Overriding

Your Answer

```
CLASS Product
    PROTECTED String productId
    PROTECTED String name
    PROTECTED Float price
    CONSTRUCTOR Product(pId, pName, pPrice)
        SET productId = pId
        SET name = pName
        SET price = pPrice
    END CONSTRUCTOR

    PUBLIC METHOD applyDiscount() RETURNS Float
        RETURN price
    END METHOD

    PUBLIC METHOD getPrice() RETURNS Float
        RETURN price
    END METHOD

    PUBLIC METHOD getName() RETURNS String
        RETURN name
    END METHOD

    PUBLIC METHOD displayProductInfo()
        PRINT "Product ID : " + productId
        PRINT "Name : " + name
        PRINT "Original Price: $" + price
    END METHOD
END CLASS

CLASS Electronics EXTENDS Product
    PRIVATE Integer warrantyPeriod // in months
    PRIVATE String brand
    PRIVATE Integer voltage // in Volts

    CONSTRUCTOR Electronics(pId, pName, pPrice, pWarranty, pBrand, pVoltage)
        CALL SUPER(pId, pName, pPrice)
        SET warrantyPeriod = pWarranty
        SET brand = pBrand
        SET voltage = pVoltage
    END CONSTRUCTOR
```

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Your Answer

```

    OVERRIDE PUBLIC METHOD displayProductInfo()
        CALL SUPER.displayProductInfo()
        PRINT "Category   : Electronics"
        PRINT "Brand      : " + brand
        PRINT "Warranty    : " + warrantyPeriod + " months"
        PRINT "Voltage     : " + voltage + "V"
        PRINT "Final Price : $" + applyDiscount()
        PRINT "-----"
    END METHOD
END CLASS

CLASS Clothing EXTENDS Product
    PRIVATE String size
    PRIVATE String fabric
    PRIVATE String gender

    CONSTRUCTOR Clothing(pId, pName, pPrice, cSize, cFabric, cGender)
        CALL SUPER(pId, pName, pPrice)
        SET size = cSize
        SET fabric = cFabric
        SET gender = cGender
    END CONSTRUCTOR

    OVERRIDE PUBLIC METHOD applyDiscount() RETURNS Float
        SET discountedPrice = price - (price * 0.20)
        RETURN discountedPrice
    END METHOD

    OVERRIDE PUBLIC METHOD displayProductInfo()
        CALL SUPER.displayProductInfo()
        PRINT "Category   : Clothing"
        PRINT "Size       : " + size
        PRINT "Fabric     : " + fabric
        PRINT "Gender     : " + gender
        PRINT "Final Price : $" + applyDiscount()
        PRINT "-----"
    END METHOD
END CLASS

CLASS Grocery EXTENDS Product
    PRIVATE String expiryDate
```

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QUESTIONS

Q1
Student Management System
[BL3 - Apply]
10 marks

Q2
2. Library Book Management
[BL3 - Apply]
10 marks

Q3
Employee Salary Calculation
[BL3 - Apply]
10 marks

Q4
Shape Area Calculator using
Polymorphism [BL3 - Apply]
10 marks

Q5
Vehicle Registration System
[BL3 - Apply]
10 marks

Q6
Bank Account Hierarchy [BL4 -
Analyze]
10 marks

Q7
Hospital Patient Record System
[BL4 - Analyze]
10 marks

Q8
E-Commerce Product Catalog
[BL4 - Analyze]
10 marks

Q9
University Staff Hierarchy [BL4 -
Analyze]
10 marks

Q10
Smart Home Device Controller
[BL4 - Analyze]
10 marks

10/10 answered

Q8 E-Commerce Product Catalog [BL4 - Analyze]

10 marks

Analyze and design a Product class hierarchy for an e-commerce system:

- Electronics — with warranty, brand, voltage
- Clothing — with size, fabric, gender
- Grocery — with expiryDate and weight

Override an applyDiscount() method for each category with different discount rules. Analyze why polymorphism makes the cart checkout process flexible and extensible. OOP Concepts: Polymorphism | Inheritance | Method Overriding

Your Answer

```

CLASS Grocery EXTENDS Product
    PRIVATE String expiryDate
    PRIVATE Float weight // in kg

    CONSTRUCTOR Grocery(pId, pName, pPrice, expDate, pWeight)
        CALL SUPER(pId, pName, pPrice)
        SET expiryDate = expDate
        SET weight = pWeight
    END CONSTRUCTOR

    OVERRIDE PUBLIC METHOD applyDiscount() RETURNS Float
        SET discountedPrice = price - (price * 0.05)
        RETURN discountedPrice
    END METHOD

    OVERRIDE PUBLIC METHOD displayProductInfo()
        CALL SUPER.displayProductInfo()
        PRINT "Category : Grocery"
        PRINT "Expiry Date : " + expiryDate
        PRINT "Weight : " + weight + " kg"
        PRINT "Final Price : $" + applyDiscount()
        PRINT "-----"
    END METHOD
END CLASS

START
    CREATE laptop = NEW Electronics("E101", "Laptop", 1000.0, 24, "Dell", 220)
    CREATE shirt = NEW Clothing("C201", "Casual Shirt", 50.0, "L", "Cotton", "Unisex")
    CREATE apples = NEW Grocery("G301", "Organic Apples", 10.0, "2026-08-01", 1.5)

    CREATE Array of Product cart = [laptop, shirt, apples]

    SET grandTotal = 0.0

    PRINT "=== E-COMMERCE CART CHECKOUT ==="
    FOR EACH item IN cart DO
        item.displayProductInfo()
        SET grandTotal = grandTotal + item.applyDiscount()
    END FOR
    
```

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[BL4 - Analyze]
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10/10 answered

Q9 University Staff Hierarchy [BL4 - Analyze]

10 marks

Analyze the design of a Staff base class extended by TeachingStaff and NonTeachingStaff. Further extend TeachingStaff into Professor and Lecturer (multilevel inheritance). For each level:

- Identify which attributes should be private vs protected
- Override calculateAllowance() at each level
- Analyze how protected access enables inheritance without full exposure. Justify the use of multilevel inheritance and explain how access modifiers enforce data hiding.

OOP Concepts: Multilevel Inheritance | Access Modifiers | Data Hiding

Your Answer

```

CLASS Staff
    PROTECTED String staffId
    PROTECTED String name
    PROTECTED Float baseSalary
    PRIVATE String ssn // Social Security / Identity Number

    CONSTRUCTOR Staff(sId, sName, sBaseSalary, sSsn)
        SET staffId = sId
        SET name = sName
        SET baseSalary = sBaseSalary
        SET ssn = sSsn
    END CONSTRUCTOR

    PUBLIC METHOD calculateAllowance() RETURNS Float
        RETURN baseSalary * 0.05 // 5% basic allowance
    END METHOD

    PUBLIC METHOD displayDetails()
        PRINT "Staff ID : " + staffId
        PRINT "Name : " + name
        PRINT "Base Salary : $" + baseSalary
    END METHOD
END CLASS

CLASS TeachingStaff EXTENDS Staff
    PROTECTED String department
    PROTECTED Integer subjectCount

    CONSTRUCTOR TeachingStaff(sId, sName, sBaseSalary, sSsn, dept, count)
        CALL SUPER(sId, sName, sBaseSalary, sSsn)
        SET department = dept
        SET subjectCount = count
    END CONSTRUCTOR

    OVERRIDE PUBLIC METHOD calculateAllowance() RETURNS Float
        SET baseAllowance = SUPER.calculateAllowance()
        RETURN baseAllowance + (subjectCount * 500.0) // Subject allowance
    END METHOD
END CLASS

CLASS NonTeachingStaff EXTENDS Staff

```



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OOP Concepts: Multilevel Inheritance | Access Modifiers | Data Hiding

Your Answer

```
CALL SUPER(sId, sName, sBaseSalary, sSsn)
SET shiftTime = shift
SET overtimeHours = ot
END CONSTRUCTOR

OVERRIDE PUBLIC METHOD calculateAllowance() RETURNS Float
    SET baseAllowance = SUPER.calculateAllowance()
    RETURN baseAllowance + (overtimeHours * 20.0) // Overtime allowance
END METHOD

END CLASS

CLASS Professor EXTENDS TeachingStaff
    PRIVATE Integer researchPapersCount
    PRIVATE Float researchGrant

    CONSTRUCTOR Professor(sId, sName, sBaseSalary, sSsn, dept, count, papers, grant)
        CALL SUPER(sId, sName, sBaseSalary, sSsn, dept, count)
        SET researchPapersCount = papers
        SET researchGrant = grant
    END CONSTRUCTOR

    OVERRIDE PUBLIC METHOD calculateAllowance() RETURNS Float
        SET teachingAllowance = SUPER.calculateAllowance()
        RETURN teachingAllowance + (researchPapersCount * 1000.0) + (researchGrant * 0.02)
    END METHOD

END CLASS

CLASS Lecturer EXTENDS TeachingStaff
    PRIVATE Integer labHours

    CONSTRUCTOR Lecturer(sId, sName, sBaseSalary, sSsn, dept, count, labs)
        CALL SUPER(sId, sName, sBaseSalary, sSsn, dept, count)
        SET labHours = labs
    END CONSTRUCTOR

    OVERRIDE PUBLIC METHOD calculateAllowance() RETURNS Float
        SET teachingAllowance = SUPER.calculateAllowance()
        RETURN teachingAllowance + (labHours * 15.0)
    END METHOD

END CLASS
```

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OOP Concepts: Multilevel Inheritance | Access Modifiers | Data Hiding

Your Answer

```
SET researchPapersCount = papers
SET researchGrant = grant
END CONSTRUCTOR

OVERRIDE PUBLIC METHOD calculateAllowance() RETURNS Float
SET teachingAllowance = SUPER.calculateAllowance()
RETURN teachingAllowance + (researchPapersCount * 1000.0) + (researchGrant * 0.02)
END METHOD
END CLASS

CLASS Lecturer EXTENDS TeachingStaff
PRIVATE Integer labHours

CONSTRUCTOR Lecturer(sid, sName, sBaseSalary, sSsn, dept, count, labs)
CALL SUPER(sid, sName, sBaseSalary, sSsn, dept, count)
SET labHours = labs
END CONSTRUCTOR

OVERRIDE PUBLIC METHOD calculateAllowance() RETURNS Float
SET teachingAllowance = SUPER.calculateAllowance()
RETURN teachingAllowance + (labHours * 15.0)
END METHOD
END CLASS

START
CREATE prof = NEW Professor("TS-01", "Dr. Alan", 80000.0, "SSN-9988", "Computer Science", 3, 12, 50000.0)

CREATE lect = NEW Lecturer("TS-02", "Sarah", 45000.0, "SSN-7766", "Mathematics", 4, 10)

CREATE staff1 = NEW NonTeachingStaff("NTS-01", "Mark", 35000.0, "SSN-3322", "Day", 15.0)

PRINT "=== UNIVERSITY STAFF ALLOWANCE CALCULATION ==="

prof.displayDetails()
PRINT "Professor Allowance: $" + prof.calculateAllowance()
PRINT "-----"

lect.displayDetails()
PRINT "Lecturer Allowance: $" + lect.calculateAllowance()
PRINT "-----"
```

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Q10

Smart Home Device Controller
[BL4 - Analyze]
10 marks

Q10 Smart Home Device Controller [BL4 - Analyze]

10 marks

Analyze and design a SmartDevice superclass with subclasses: SmartLight, SmartThermostat, and SmartLock. Override an executeCommand(cmd) method in each, where the same command (e.g., 'on', 'off', 'status') produces device-specific behavior. Analyze:

- How polymorphism allows a single controller loop to manage all devices
- Security implications of data hiding in SmartLock (PIN must never be exposed) Design the class hierarchy and explain the OOP principles applied at each level.

OOP Concepts: Polymorphism | Encapsulation | Inheritance

Your Answer

```
START
class homedeviccontroller
  declare device name as string
  declare status as string
  function turnon()
    status="on"
    display devicename + " is turned on "
  end function
  function turnoff()
    status=off
    display devicename + "is turned off"
  end function
end class
main
  create object device of homedeviccontroller
  input devicename
  device.status="off"
  display "1.turn on"
  display "2.turn off"
  input choice
  if choice=1 then
    call device.turnon()
  elseif choice=2 then
    call device.turnoff()
  else
    display "invalid choice"
  end if
stop
```

[Save Answer](#)

10/10 answered